ActInf Textbook Group ~ Cohort 5 ~ Session 4 (Chapter 2, Part 1) ~ 10/10/2023

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all right welcome back everyone it is uh October 10th 23 and we're in our first discussion on chapter two so before we jump into any questions or anything what is anyone's just takeaway from chapter 2 or what were they curious about or what something that kind of remains with them after reading feel free to just raise your hand hand or just go for it yeah hi it's Pablo here um in the page 35 action has a pic value and pragmatic uh value um brings Rewards so that for me for what I'm building is super important so yeah I I have also some other notes but and a question over there we can cover later so yeah for me this is like my pick on this awesome yes the idea that we don't have to coers epistemic value or learning or Information Gain into utility we don't have to like convert it into utility in order for that policy to be selected key aspect any other favorite quotes or fun parts of chapter two or parts they woke up sweating about or I for me tying you tying this back to the surprise and and and um maybe elaborating on what encompasses policy I I'm I'm putting strategy in there but I don't know if that's correct specifically in the low road cool like policy enumerates the possibilities for Action over a given time Horizon but strategy isn't just enumerating branching path so then what is strategy great questions any other just random chapter two thoughts you know when it comes to this idea of planning as inference something that I've been just kind of thinking about a lot lately is that um ultimately when I do something and the best example is just trying to walk somewhere all I really do is to hold that conscious intention to be somewhere and then then I'm kind of out of the loop other than having uh you know processing um sensory evidence that I'm actually getting there and as long as long as that's happening you know I I can sort of see this this Loop in my in my brain where I'm going and just enjoying myself looking at things and just thinking all kinds of other thoughts and so on so forth and the going kind of happens on its own all all I really do is to have a conscious intention I don't even necessarily have to feel my body as it's moving if I get into this state almost like a you know like a walking meditation then it really feels like you know being in a self-driving car so it's kind of interesting that um you know I'm starting by understanding all of this stuff like having a having a generative model to notice these things in my own awareness I'm starting to notice these things in my own awareness and I'm starting to be you know not only uh have have these ideas confirmed mathematically but also exponentially so just a little bit weird but um really fun awesome cool and and also no amount of propositional intention to walk you can think as hard as you want I want to walk but then actually the intention that gets you out of the chair is something different so what does that what does that mean or what does that have to do with action yeah and but but but so you have to you have to sort of um distinguish different kinds of intention so like there's this there's the intention as in kind of like it would be nice if or like wishful thinking I I I wish I could do this um but what I mean by intention is a very specific mental state and that mental state is usually not available to you unless you start training in meditation where it becomes obvious that it exists so it's something that like you know if if you think about like think about this in from the perspective of training neural network for robot to do something so you know robot will have some kind of goal in a neural network getting to a place or whatever so it must it must be some kind of activation in the neural network that's that's going to be triggering a sequence of other activations right or a bunch of activations that are going to go through some kind of cycle in time that that sort of thing and normally that's an unconscious thing but I I don't I don't think it has to it has to be so and like the Buddhist meditative practices particularly inside meditation they specifically you know distinguish the intention so at some point you might be become aware like first they have an intention to move the hand then they have a feeling of moving the hand so as the I'm holding the intention and I'm getting the sensor input to confirm that what I intended is happening and that process is actually uh it's possible to introspect it's possible to to see it in your own awareness and that's um that's that's wonderful really awesome any other opening thoughts from people just some low road or chapter two thought or something that they're excited about to explore a little bit today um the um the getting a little deeper into what is learning versus inference yeah multiple time scales that computational models of cognition play out on the fastest sometimes being called inference the kind of Rapid Fire neural spiking type little bit slower attention with a Precision waiting stepping on the gas pulling off the gas and then a little bit slower even learning sometimes the whole process is called inference sometimes inference is only being used to refer to the kind of like crack of the whip most rapid time scale but what is learning the higher order phenomena like child learning how to read how do we connect that with a more narrow statistical concept of learning like we leared that the average height of the tree in the forest was 10 feet but previously we thought it was nine so that's learning so is that the same as belief updating what would anyone else say what's what is the relationship between learning and belief updating I'm curious about um if uh this system is uh deterministic or it's more on the will uh Free Will side and or or nonone at [Laughter] all are you say saying about the belief updating or about the active inference about the active inference okay you know I I like to joke that free will you know you have you have free will only so far as that you're free to suffer if you really understand that you're not a thing that you're a process and you're totally interdependent with everything else then you know the um the best thing to do is always to you know be a free energy minimizing agent to do it in a in an optimal way if you're not doing that in an optimal way that's cause that will cause some friction or suffering we can jump to this uh cohort five question that's kind of related um and then yeah go who was that okay um sorry oh yeah Ali go for it uh okay so uh the relation between belief updating and learning is that belief updating is much more General than just learning so uh both inference and learning um involve belief updating in some sense and uh this is uh one of the probably most confusing things about active inference modeling because most people when it comes toar learning they only think of belief updating uh as apply as applied to Only The Learning side of active inference uh but uh then basian belief updating uh can be applied to both inference side and uh the learning side so they use the exact same kind of belief basian belief updating mechanism uh so uh we'll see how Works uh in relation to both more explicitly chapter six yeah the faster inference scale is kind of like you have the calculator preloaded like the weights in the neuron Network are fixed if you want to think of it that way or the parameters are kind of like pre-loaded and preset and then inference is like new argument comes in new input comes in and it just it's like a plug-in chug that's the faster um inferential scale which can still result in a belief updating and then learning is actually the change on those weights or like the change in that in that system how it's set up so then influences future rounds of fast inference um so in that respect um it's almost like what Kristoff was saying uh Visa inent ality prior to a generative model where learning is changing those parameters you know prior to having the generative model being able to update that intentionality which the which then the belief updating happens in relation to perhaps learning could be like what kind of thing am I and then the faster inference is like given the thing I am how do I respond and then the kind of General theme is instead of evaluating sense making or decision making in terms of like maximally rewarding their cast in terms of most likely so instead of given what I value what's the most rewarding it's like given the generative model what's the most likely outcome and then in chapter nine we'll look at at given what empirically did happen what's the most likely generative model so that's the kind of scientific like behavioral researcher going from the data to the most likely model and then the other direction is from the model generating the most likely data and both of them are are um they're just two two directions on the same freeway um all right let's look at this question and then anyone else can like add a question to the table or or um write another question in the um chat okay this means that Free Will is part of the model question Point 2.3 the interplay of top down and bottom up processes distinguishes the inferential view from alternative approaches that only consider bottomup processes okay probably a lot that um could be explored here what what what does anyone want to kind of bring in here or say about this section yeah that that was my question so yeah my thought was um if it's Boton up it would be determin everything is around U it's um getting like uh um you are just like uh observing and just getting the data in and you probably won't have much um action on on that or your agency I think we call it and if you is top down it means that I have an agency I'm deciding I'm I'm modeling the the world that surrounds me and um yeah that was my my question my thought and something that I care it's very pH philosophical as well I think and and yeah anyone have a thought on this yeah this is Andrew um I when I hear this question I I just want to recommend Eric hell's new book um I I forget what it's called right now um sorry I'm multitasking as usual um but he he talks about how those uh I think active in inference um is completely congruent with his theory of um causal emergence which talks about how you can have bottom up processes and top down and and still have some room for free will in between um with just uh mechanistic um uh rule folling behavior and and so I feel like there's room for um the philosophy to still exist Within a um a a uh non-dualist uh approach to um metaphysics yeah thank you Andrew yeah yeah thank you that makes sense if I may add a little bit um you know I I think it's interesting to to consider here um sort of Steven wal's principle of computational equivalence so um W from based on his experience with computation says that essentially um there is a level in the computational ladder ladder of sophistication and it's actually pretty low it's it's incredibly low that um once you hit it you're as sophisticated computationally as any other agent so even if you had a completely computationally deterministic system where everything is fixed like the universe there is absolutely no way you're going to be able to properly predict um out compute um even the simplest agents like simplest cellular autom and so on because they are effectively as computationally sophisticated as you are so that gives always rise to um uncertainty it's it's a building feature that uh you will not be able to compute the perfectly a model of internal

States and this is comes up in active inference modeling the difficulty of constructing these generative models but it seems a fundamental principle of nature that that's impossible so as an agent you know um if you're as an agent you're kind of a controller for future State you have an intention that you want to achieve a future State and you're trying trying to sort of get there but the world that you're trying to control is never going going to be um simply deterministically following rules so there's going to be resolution of some uncertainties and I think that resolution of uncertainties that gives the rise to the feeling of of free will the underlying system might be completely deterministic but the experience of it from from within an um a part of it embedded in it in the system it's always we can to feel that I had these choices because I always had some uncertainty yeah yeah yeah that that's another thing that I Steve W from it's I I always wonder how those these two currents are going to cross over yeah yeah I suspect on a machine learning Street talk with Carl and Steven in the coming months we'll see though that should be quite um interesting um Oli and then Susan uh okay so when it comes to agency I suppose active inference kind of assumes that any lifelike property including uh agency and um many other things uh they're kind of uh inherent to uh and uh also their emerging property of any dynamical non-equil Dre system uh and most of more often than not these systems are erotic and also they possess markup blankets so um uh in other in other words F supposes organisms have this uh goal of keeping themselves in their uh expected States uh or more precisely their expected phenotypic and ontogenetic states and uh consequently they act to minimize the discrepancy between um the desired or predicted the State of Affairs uh and uh the one that they experience so uh uh to Cote frisen himself uh it furnishes an account of basic or biotic sentient behavior that places agency Center Stage so uh I believe active inference and Fe uh is not a deterministic uh theory of cognition at all uh because uh as I just quoted friston it takes agency very seriously and uh even at the very cental stage of the theory thank you Oli Susan I kind of to toe on that and you and pose a question in terms of agency um I you know I kind of question whether we Rec we recognize without learning our requisite limits which is kind of what you I interpreted what you were talking about um you know people go over their requisite limits all the time it's called anxiety it's called you know schizophrenia whatever but but it's this has kind of got me thinking that the agency and the ordance is is kind of a chicken and egg problem it's like you know as I as I update um is I find more freedom um which you know freedom to choose and in fact you know when I heard the intention I was like well I have a choice of other intentions so that's going to Cascade down um that's you know that's basically agency but but I still have to so the belief updating or learning um you know is going to show me new affordances but then I need new agency to be able to do something else with that affordance so you know so is it kind of a chicken and egg problem or how does that play out in this model a few narrow points but first Ali please uh well yes actually uh there is some concept overlap uh the FP uh with the kind of ecological psychology uh Viewpoint offered by Walsh in terms of affordances so uh Walsh argues that uh if an agent has a goal it will experience its environment as a kind of menu of affordances uh what's in its surroundings and will and uh whether uh they're useful or not for attaining that particular goal so uh the environment agent interaction might might be regarded as a kind of landscape uh that uh that should be navigated to that given end uh so in effect what it says is uh it it posits a kind of a optimization problem uh in which um one might effect efficiency um uh to play a role in determining that trajectory so that's basically uh the overlapping between the formulation of Fe and uh this uh psych ecological psychology Viewpoint or at least Walsh's formulation of eonian affordances because uh in both of those um cases uh there there's this bilateral uh interaction between um between the environment and the agent so as to uh the chicken egg problem uh it cannot really be separated from each other so uh there's this continual um a bilateral interaction between those two Co yeah just just to narrower points so Pablo kind of kicked the soccer ball off with this question about bottom up and top down and uh these generative models are as constructed so we could imagine a top down constraining Force that's just always doing it's always clamping the same way or it's alternating between two different top down impositions so just the the the mere presence of top down forces does not support or reject in the system of Interest itself the presence or absence of Free Will defined as as someone chooses um but by comparing a portfolio of models it might be possible to to Garner evidence or kind of um come to favor one of those choices more and similarly the um information coming bottom up could be generated deterministically it could be an oscillator or it could be generated probabilistically like being drawn from a distribution um and that could drive belief updating in the agent in a nonlinear way or in a linear way so it's like sometimes these um adjectives that modify the probabilistic intern jals of the model deterministic probabilistic all of that um there's there's a lot of richness but also openness in how they articulate with um the broader philosophical questions like Robert spolsky's current book or it's coming out in some days or something like that it's called determined the science of behavior or living without Free Will so it's like deterministic sounds like it's going to be addressing exact same question that spolski is going to be addressing but then as Kristoff pointed to even a deterministic system of quite limited sophistication the only way to predict Its Behavior may be to play it out this is kind of like turing's halting problem like even if you're an expert in programming you look at a piece of code you may not be able to know a priori whether it's going to Halt or how it's going to act um and then again to this kind of like more limited generative models have a zone of surprisal around them but the zone of surprisal with respect to the modeler isn't the same thing as a truth Claim about the system you could make um an example where the affordance landscape is fixed like the rat in the teamas that we'll see later but a real rat in a te-as might discover a new affordance like it might find out it could roll over and you know knock something over and then if it has that capacity to learn that could be model as like expanding its e variable like expanding its um repertoire of affordances and then it might come to select those so then it but the fact that a given generative model does or doesn't have the ability to expand its repertoire of affordances the real rat on the ground just because the model can't doesn't mean the rat can't just because the model does doesn't mean the rat will so that's the kind of iterated continued engagement and repeated modeling as a way to develop hypotheses to to reenter the system of Interest with action not to just make version one of the generative model shut the door on the question cool well what's another question or section that somebody wants to look to I'd really like to capture what you just said about the rat that was Pro prolific oh mammals um let's look at some classic questions there's a lot here but does anyone want to just ask one just that comes to mind now or something they scribbled in a margin um yeah inks and then oh yeah inks first and then Andrew y so considering the difference between model and process I'm still very far behind all of you guys so I was thinking the model doesn't have to be true right it's subjective it has to work it doesn't have to be true does the process have some limit like do we know absolute truth on the X asterisk the hidden state that is asterisk as part of process I'm thinking that connecting that to um I don't know if you guys have read Greg's Egan uh Greg Egan's Shields ladder where they discover a v well by mistake through an experiment they trigger a vacuum that is a more stable State than our vacuum and it starts unrolling through the universe the way Ice Nine does in cat's cradle by Kurds kurd vonet so I was thinking our mod like basically inventing more and more stable models they roll out like that like a carpet through everything because they're so much easier to survive and persist through but that's on the process side the real real hidden state does it have truth yeah that that reminds me of this very short borus um story where um you know this is this is a somebody read chapter 6 little bit too literally um and they wanted a perfect map and then it it came to just like you said inks it came to unfurl over the whole territory until it just blurred with with the territory another answer to that is um xstar the the generative process using the way that the textbook talks about it though Oli may may bring up a very Salient point is um still something that's proposed by the model and so xstar could be said to have like a local truth let's just say we have um you know an air condition turning on or off agent kind of regular setting like that thermometer sense readings making inference about temperature in the room taking action to turn on the air conditioner and then we have like the generative process which is giving rise to the temperature readings now all of that is like a local constructed truth but just making some sort of function to describe how temperature um plays out in that um modeled room again that doesn't mean that the real room has that feature Oli uh well yes there so there's actually this um kind of criticism or uh debate going on around FP um about whether it's uh too General to be uh realistic uh scientific model of uh the agents or more specifically cognitive agents because on one hand um it kind of puts everything in a a great degree of abstraction uh to the extent that it may not be able to answer some uh some concrete um questions about the biological or the bi olical nature of the uh agent or the real world situations of the agent so for instance uh there was this a paper by uh coloman palasio from uh 2020 one uh namely uh let me yeah non-equilibrium thermodynamics in the free energy principle in biology uh in which um they explicitly put forward this criticism around FP as being to too General to be useful for modeling any real world or or concrete uh biological problems but on the other hand uh FP never claimed otherwise because uh according to the fe uh any or a given non-equilibrium system uh with a Target State um will all would always seek the most efficient route uh and this route would typically uh be the one defined by considerations of uh the gradient descent or in other words the path uh through all the available configuration space um uh that follows the uh steepest route uh to the minimum so uh again in this view uh it's it's not even uh explicit whether the agency or all the other Concepts we're talking about are real or or the apparent properties of the agents uh so um more to the point uh I believe uh Fe and active inference can be more congruent with a structural realism uh view of the uh philosophy of science as opposed to just um old-fashioned or traditional uh real pure realism or naive realism of U view of the science uh and in this regard uh Frist coauthored some papers with M Majid Benny uh and they argue for this view of f as uh as a kind of view that that is more congruent with a structural realism and particularly ontological uh structural realism awesome Kristoff you know I just want to say that um holding on to any kind of notion of Truth U I think the the best the best thing that ever happened to me in my life is I decided that just there is there are no truths there are just simply models within models within models and you can sort of see it in in the temperature example temperature is not real it's a virtual variable literally what what the physicist would call it right it's a um it's a way of cor spraining all the um interactions between gas molecules in such a way that we that we can measure so we have something that persists for time that we we can observe but you're you're never really looking at reality and if you if you go down you know this rabbit hole then ultimately you will arrive at something that like there is some computational say W's perspective on this am very fond of that one there is some computational model that's running all possible computations so what truth would be in that in that kind of context would be that the the the shortest description um the shortest possible program that gives rise eventually to what what you're observing but because un because Universe because everything is kind of like interdependent you're never going to be able to to to to I think I think you're never going to be able to separate um you know and say like oh this is this is the the the the the the true process that gave rise to to to this observation it's going to be some kind of interplay of all kinds of computational process all all running kind of independently so um you know in so far as I think this is why there there's this difficulty in in writing down these generative models of all these possible um Poss when we have this um you know probability of of our priors because there's absolutely no way to to to know what these priors will be we can only make um uh you know guesses of what the what the range of these um uh values uh can taken and adapt as as um evidence evidence as observations are flowing in cool also on this kind of generative model generative process uh point this is one great great question um in figure 22 and throughout this textbook pains are made to distinguish the generative model from the generative process and it is a useful distinction to understand the process that gives rise to the observations the modeled process that gives rise to the observation so the kind of like digital niche of this generative model of the agent um and yet in a guest stream we saw a slight different perspective or at least way to talk about this when Maxwell presented and utilized this bracket to cover the entire generative model including external States again those that give rise to sensory States and at this time step in the published transcript Ali asks this question very specifically so there's of course many ways to speak this or any dialect but within this textbook generative process is used to refer to the process external to the agent the particular states of the agent blanket and internal the external states are described as generative process giving rise to the observations and not influencing the actual fundamentals of the particular partition but rather just using a different natural language to describe it it is also becoming common to describe the generative model as describing all the necessary and sufficient statements that the modeler makes about the entire scenario and not distinguishing so much as to what's the generative process in the generative model one piece of uh I guess support for like this kind of just use generative model to describe it all first off it's simpler also if you have two agents in conversation then of course one of them is the generative process to the other and vice versa and so just In The Same Spirit of like well states of the world or aspects of the world let alone the map they're not tagged as like sensory or action or internal or external those qualifiers are always relative to which node is being called the internal State and then the rest of the blanket is just trivially assigned from there so it doesn't make sense to like say well this desk is the external State it's like well but it's its own internal state if I chose to model it from that side or with respect to the wall behind it it's the blanket State between me and the wall so using generative Model A little bit more broadly seems to be a little bit of a development that has unfolded over the uh cohor and years Susan yeah I mean the way ask see it if I you correct me if I'm not looking at it correctly but you know when you do research you have to decide what your ontology or epistemology is going to be when you apply it um and um and you always provide context for that so this kind of is above that in my mind in fact you know we and you and I've had a conversation because I yeah I want to make sure that the that it's um that it's ideology free um because you know I see way too much um you know um BS especially in management science that you know is just embedded with somebody's ideology and it's taken as fact so it's kind of what I like about the model um that um brings me to um you the the marov blanket and um you where you so in that sense of you the uh the model the generative model where where does the Markoff blanket extend because it was a little confusing reading it is that just confirm for me though am I looking at it correctly in terms of it being a theistic that's why I posted that in the comments because that's what I copied from chapter 10 I think cool maybe post what what page you you found it on but it's it's a great quote um features are not intrinsically on the blanket or internal or external there's a base graph that conveys whatever variables are being included whatever Quantified aspects are under consideration or or formalized doesn't have to be a number it could be like categorical but whatever is being brought whatever Lego pieces are coming on the table and how those Lego pieces are connected that's the base graph and then whatever node or cluster of nodes is being selected as the internal States the blanket is just all that surround that and so when we're taking this kind of agentic view or cybernetic view on cognitive ecosystems then the blanketing states just to give a first pass can be understood in terms of sensory data so anything that you would record with a sensor and then the outgoing dependencies are actions anything that would be described by the affordances of an actuator so to add so to add build on what Ink's question was would the Markoff blanket be the same for if I'm doing let's say I am doing a research in a modeling project would I use the same Markoff blanket for the process and the model or would the model or would the process change the blanket the the final mod just giving a narrower answer because I I I know that's something that could be explored at Great depth if you made a generative model with a fixed Mark marov blanket then it would be fixed if you said our room has a thermometer and an arm that's an actuator and you know different data are going to come in and the arm is going to do different things well then the the kind of um scheme of the blanket wouldn't change but you might want to be interested modeling the transition from that room deciding that it should put out a second thermometer at which point the scheme of it would change okay and so then you'd be outside that room be like oh they literally just went from having one thermometer to two thermometers and then you could postulate cognitive models that happened inside of that room you'd say maybe it's like there they had an affordance to add a thermometer and they selected that affordance does that answer your question better in now I have more more questions but I don't want to stir more trouble here what what else is the end of a session for other than stirring trouble um so because you were talking about being in the room can I just roll with it of course you were talking about being in the room and then a second th thermometer and I was thinking of as an agent this I don't think this is about this chapter but as an agent as my mark of blanket is like I'm looking at the screen right now but then have a flutter in my heart and I'm like oh no am I having a heart attack in my mark of blanket is that a separate Mark of blanket where now I'm modeling myself very close and tight around my heart then I hear the door knock and oh I have a package from Ukraine and I run like now my My atten sl Mark of blanket is all the way my apartment and the door to the outside world where someone is knocking are these different am I switching between different Mark of blankets or is the mark of blanket moving with my attention regime great question well to the wandering blankets and um blankets that are not sort of one or zero but can be on a Continuum or a blanket index certainly some of the most recent work is addressing that I'll just leave that there but to the interception question um in chapter six when we're exploring ing how to construct these generative models an example of of Sensor Fusion is provided so let's just say that a given body has some interceptive Sensations like heart rate and then some exteroceptive sensory capacities like hearing and vision and so on um those are kind of like all open even when you close your eyes you're still getting visual input and and usually it's modeled not as changes in the blanket but in changes in attention like you still have proprio receptors in your foot but you're not paying attention to them you're not attending to them which is to say it's as if this the data coming from them are not making a difference to belief updating if you're not paying attention to something it is not having consequence it's its differences are not making a difference and then regimes of ention describe the way that in this kind of multimodal setting um the spotlight of attention might focus on a given sensory modality and so that is usually modeled more as an attention modulation task within a fixed marov blanket schema rather than um changes to the the schem of the blanket itself Kristoff so you know this makes me think that um having this sort of uh you know explicit model of attention within um a fixed Mark of blanket it's really it's really Mark of blanket is a kind of like a convenient um it's it's an idea it's a convenient idea for modeling to to make this separation but it's not actually something that that is real there is um any boundary that you can find and say like yes this is definitely a boundary between the internal and external States and this reent work on on on on on moving blankets I think I think what we're eventually going to realize is that it's there's just a Precision parameter there as well like how how how how thick I want my my market blanket to be and um it's for for the convenience of describing certain phenomena at certain temporal scales it simply makes it makes sense to make an assumption that there is a separation between these internal and external States but in reality if you look at it you know at all temporal scales you will find that there is the the the separation is you know it's not really there it's a it's a it's um it's it's invented so this you know it's kind of like uh this is the way I sort of understood the the the Buddhist concept of relativity everything is sort of relative and I I put myself in a frame of reference and I said well within this frame of reference these are internal these are external States and then I have and and and I have a way of of looking at them to make sense of them if if I didn't have the ability then it might be difficult to to be able to make sense sense of the world if I if I if I can't make this partition but is the partition real I I you know personally I didn't think so it's just it's just a point of view thank you um Andrew yeah a couple of riffs on um what's been said uh one like the the the finding your own Markov blanket I feel like is a little bit of a um hopeless cause because it it's it's very likely that you are many nested um you contain Markoff blankets all the way down um and there's um a lot of complex phenomena and interior modeling going on um in terms of coming to grips with that one resource I'd like to recommend is if anyone's seen the book The Mind illuminated by Dr John Yates um it's it's kind of a good rationalist take on a meditation practice that uh kind of lets you come to grips with there's probably multiple agentic things happening in your brain um another thing that uh was just kind of wow to me that I just wanted to gush about about this chapter is um I've been trying to sort of come to grips with what normativity means in a um sort of non-dist agentic um frame of reference and there was a couple quotes at the end of this chapter one of which is in active inference the identity of an agent is isomorphic with its priors that just like I was like oh all right that kind of puts it all together and um kind of um to to get that um pitched up they start with expected states are preferred and include the organism's conditions for survival EG Niche specific goal States whereas their opposite surprising states are dispreferred in this way by find fulfilling their expectations active inference agents ensure their own Survival so it just that phenomena one thing um your Beijing priors are your normative order which um I've been searching for what does a normative order mean especially within the context um sort of a small agent um trying to find relevance in in the cognitive limits of finding relevance there and um I feel like this is um this is the right approach also one more thing with the um free uh not variational free energy but um um the the planning one expected free energy um we were talking about um what s the way it includes both epistemic and um preference value in the same equation I I feel like as a group we really need to um tackle that as a and realizing that maybe a lot of our preconceptions that we've been grappling with during this meeting are really summed up with understanding that itself that that it all fits both epistemic ontological in the same sort of optimization framework so that's that's just me um riffing on this great comments equation 2.5 and equation 2.6 are like the heart of the heart of active inference we have the real time sense making about incoming sense data and beliefs and then we have the prospective which not all agents even need to be modeled as doing the prospective unified imperative for pragmatic and epistemic outcomes and that's the heart and that's the mystery and that's um like the part that we don't need to reinvent and then all the work is in getting it into a form getting the system of Interest into a form where you just press play on those equations um Ali and then any final comments that people have thank you uh yeah uh regarding whether or not or to what extent markup blankets are code UN code real uh it's probably worth uh keeping in mind uh that markup blankets are uh more precisely they're not necessarily spao temporal boundaries but rather uh they draw boundaries in state space or in other words in the space of all possible configurations uh so uh it's true that sometimes it manifests itself as a kind of spatial temporal boundary such as uh the cell membrane uh but uh more often than not uh we don't need to consider markup blankets as uh kind of purely physical or spatio temporal boundaries so uh yes it's a kind of uh modeling technique uh but but uh in terms of its mathematical um reality it is as true as any other mathematical models cool well very fun the low road is it a bumpy road is it a smooth Road what is it like I don't know um but that was our first discussion on it next week at the earlier time we will check back so it would be epic if people just right in the coming minutes or in the coming week add a few more questions to chapter five questions where we'll look at the bigger questions table too um in cohort 4 we'll be heading into chapter eight on continuous time generative models and uh we'll have next week on chapter two then head into three so thank you all the game goes on so see you next time thanks everyone bye bye